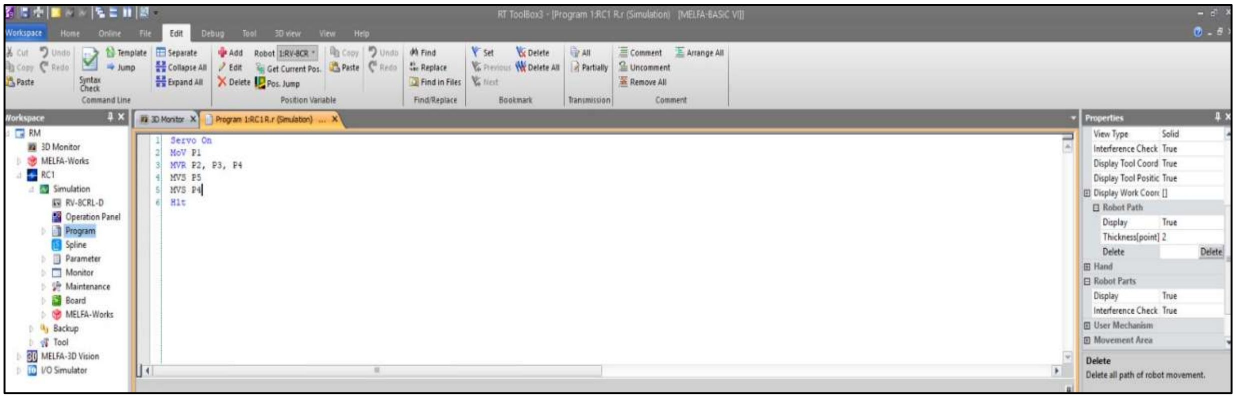

Figure 1: Source Code for Robotic Interpolation and Trajectory Planning (R Letter).

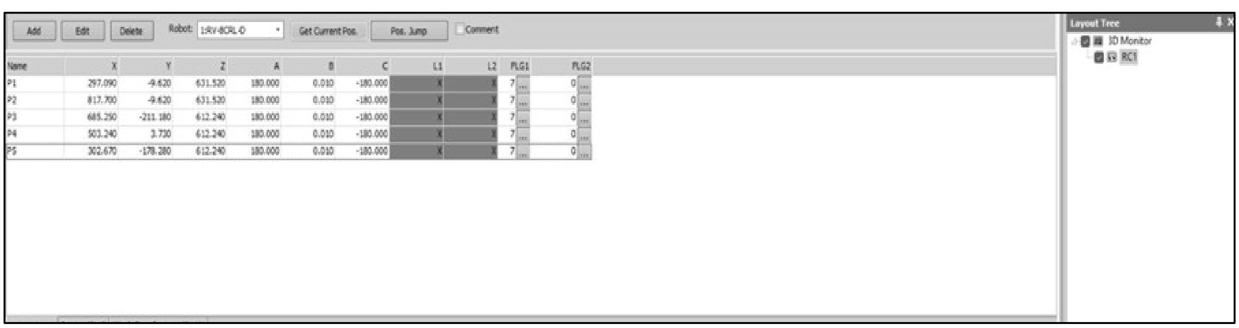

Figure 2: Defined Workspace Coordinates for Trajectory Synthesis.

Expt No:
01
Date:
28 / 01 /2026

ALPHANUMERIC AND SYMBOLIC TRAJECTORY SYNTHESIS

1. AIM:

To design and implement a high-precision robotic trajectory for the synthesis of alphanumeric or symbolic contours through the orchestration of linear and circular interpolation techniques on a 6-axis industrial manipulator.

2. APPARATUS REQUIRED:

a) Manipulator System: Mitsubishi RV-8CRL-D 6-axis industrial robot.
b) Control Hardware: CR800-D series dedicated robot controller.
c) Software Suite: RT ToolBox3 integrated engineering environment.
d) End-of-Arm Tooling : Precision drawing effector configured within the tool coordinate system.
e) Infrastructure: Solid installation platform and standard TCP/IP or standard USB communication interface.

3. THEORY:

Trajectory generation for planar symbolic mapping requires the mathematical decomposition of characters into primitive geometric segments.

a) Core Principle:

The robot Tool Center Point (TCP) is commanded to follow a continuous path synthesized from individual motion commands that dictate both spatial orientation and execution velocity.

b) Working Mechanism:

i) Linear Interpolation (Mvs): The system calculates a straight-line motion vector between two spatial coordinates, maintaining constant velocity. This is essential for rendering the rigid segments of characters (e.g., 'L', 'H').

ii) Circular Interpolation (Mvc / Mvr): Utilizing a three-point reference logic (Start, Auxiliary/Passing, and End), the controller mathematically generates the rotational arcs required for curved characters (e.g., 'O', 'S').

iii) Continuous Path Control (Cnt): By invoking Cnt 1, the controller bypasses deceleration at intermediate via-points. This ensures fluid motion blending across trajectory nodes, thereby preserving geometric integrity and minimizing cycle time.

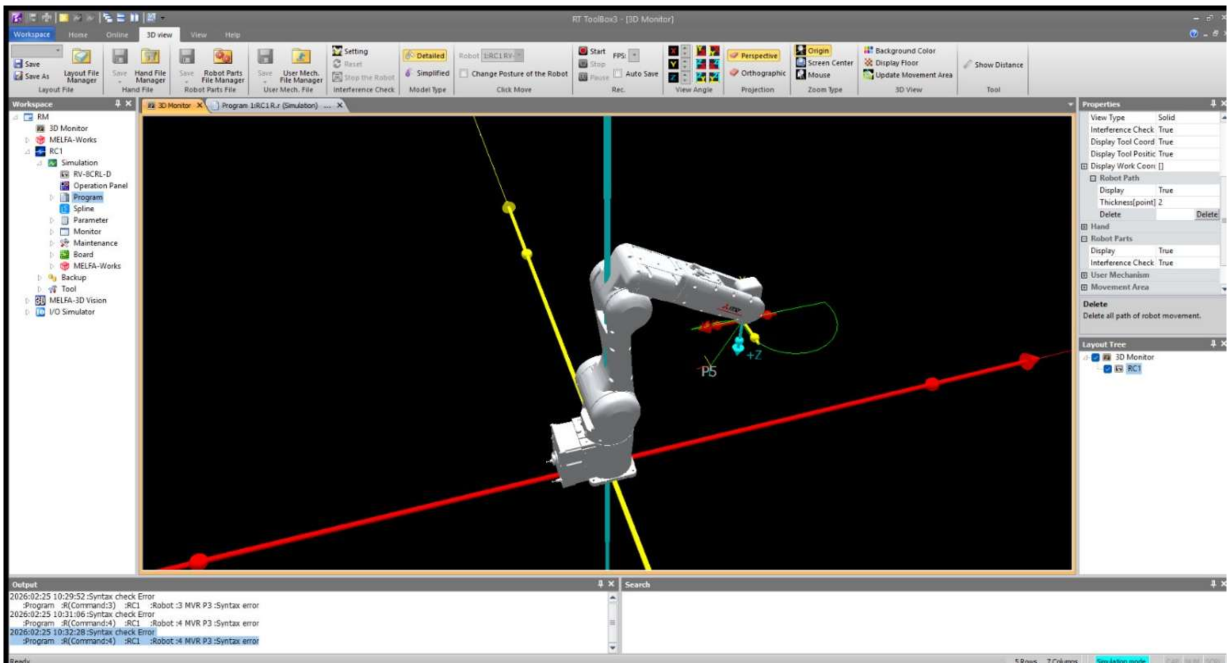

Figure 3: Simulated Kinematic Trajectory in the RT ToolBox3 3D Monitor**.**

4. PROCEDURE:

a) Setup & Connections:

Establish the communication link between the engineering workstation and the CR800-D controller. Perform an origin data verification to ensure absolute axis calibration.

b) Algorithm & Logic:

i) Initialize servos and designate the drawing tool parameters.

ii) Execute a joint-interpolated move to a stabilized approach height directly above the initial character coordinate.

iii) Sequentially trigger interpolation commands (Mvs for straight vectors, Mvc for curves) to construct the symbol.

iv) Incorporate temporal delays (Dly) between active strokes to allow for mechanical vibration settling.

v) Terminate the cycle by retreating to a verified safe home position (phome).

c) Execution Steps:

i) Load the synthesized logic into the RT ToolBox3 program editor.

ii) Perform a Step Run in manual mode at reduced override speed to validate the path and ensure collision avoidance.

iii) Transition the controller to Automatic mode for high-speed trajectory execution.

d) Observation Instructions:

Utilize the RT ToolBox3 Oscillograph tool to analyze motor current commands and axis load levels, specifically noting load spikes during high-speed curve transitions.

5. RESULT AND OBSERVATION:

The system achieved accurate contour synthesis with smooth motion and consistent precision throughout the operation, as verified by faculty evaluation.